

Fragile Watermarking Algorithm for DCT-Domain Image Authentication and Recompression

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Abstract

A compression-watermarking algorithm which embeds a watermark bit by modifying the last coefficient in each quantized DCT block is proposed. The algorithm provides both authentication and recompression capabilities. The simulation results show that the proposed scheme successfully detects and localizes tampering modifications and improves the coding efficiency by approximately 5%.

1 Introduction

Fragile watermarking [1-3] is designed to enable the authentication of multimedia content. Traditional watermarking algorithms satisfy the general constraints [4] regarding embedding information into multimedia, but take no account of the coding efficiency. Accordingly, this paper proposes a novel compression-watermarking algorithm in the discrete cosine transform (DCT) domain which not only provides a watermarking function, but also enhances the compression ratio of the watermarked multimedia content.

2 Proposed compression-watermarking algorithm

The aim of the proposed compression-watermarking algorithm is to insert a watermark bit, w , into each 8×8 quantized DCT block, q . The algorithm commences by applying a run-length selection strategy to select the specific DCT coefficient in q at which to embed the watermark bit. The run-length selection strategy takes into account the requirements for coding efficiency, perceptual transparency and sensitivity to tampering attack. These requirements are satisfied by selecting the last non-zero coefficient in the block (as identified in a zigzag scan) as the coefficient at which to embed the watermark bit.

Let this coefficient be denoted by φ . In embedding the watermark, φ is modulated to $\bar{\varphi}$ using Eq. (1), resulting in a watermarked block, q^w , i.e.

$$\bar{\varphi} = \begin{cases} S(\varphi)(|\varphi| - 1) & \text{if } w \oplus E(\Lambda) = 0 \\ \varphi & \text{if } w \oplus E(\Lambda) = 1 \end{cases} \quad (1)$$

where $S(\varphi)$ indicates whether φ is positive or negative (the number's sign), and $|\varphi|$ denotes the absolute value of φ ; $E(\Lambda) = (\Lambda + 1) \bmod 2$, and Λ is the number of odd value coefficients in q .

The watermark extraction procedure is equally straightforward. The extracted watermark bit, $\bar{w}$, is determined simply by assessing the total number of odd coefficients in the watermarked block, i.e.

$$\bar{w} = \begin{cases} 0 & \text{if } E(\bar{q}^w) = 1 \\ 1 & \text{if } E(\bar{q}^w) = 0 \end{cases}, \quad (2)$$

where $\bar{q}^w$ is the reconstructed watermarked block.

Implementation in JPEG domain: In this letter, the proposed algorithm is applied to the JPEG coefficient domain to evaluate its watermarking capabilities. Figure 1 illustrates the proposed watermark insertion procedure. Assume that an 8-bit grey-scale image I of size 512×512 is to be protected by the insertion of a binary watermark W of size 64×64 . The number of watermark bits required is equal to the number of quantized 8×8 DCT blocks in the image. In the watermark embedding process, the watermark information, W , is permuted as W_p using a secret key, K , as the seed of random number generation. 8×8 DCT transformation is then performed sequentially on the original image I , and watermark bits are embedded in the corresponding 8×8 quantized DCT blocks in accordance with Eq. (1). Finally, variable length encoding (VLE) and entropy coding is applied to the resulting DCT coefficients

to produce the watermarked image, I_w .

Referring to Fig. 2, an assumption is made that an embedded bit “0” is to be inserted into the DCT block. Since the total number of odd value coefficients in this DCT block is 3, i.e. $\Lambda = 3$, then $E(3) = (3+1) \bmod 2 = 0$. In accordance with the run-length selection strategy, the last non-zero coefficient in the block is selected as the location at which to insert the watermark information (see Fig. 2). From Eq. (1), the value of this coefficient is modified from 1 to 0. In this particular example, the watermarking process reduces the total number of run-length symbols from six [(0,56), (0,12), (0,7), (0,3), (0,4), (1,1)] to five [(0,56), (0,12), (0,7), (0,3), (0,4)]. Therefore, less coding information is required to encode the run-length symbols, and hence an improvement is obtained in the data compression ratio.

In the watermark extraction process, shown in Fig. 3, the input image, $\tilde{I}_w$, is initially decoded using variable length decoding (VLD) to obtain the watermarked blocks. The watermark bit is then extracted sequentially from each watermarked block using Eq. (2). The extracted bits are assembled to generate a binary image W'_p , which is then inversely randomly permuted using the secret key (K) to extract the watermark, $\bar{W}$.

Consider the case where the watermarked image has not been changed, e.g. as the result of transmission errors or through tampering by a pirate, i.e. $\tilde{I}^w = I^w$. In this case, all of the reconstructed quantized DCT blocks are identical to the original quantized DCT blocks, and hence $W'_p = W_p$. Also, assuming that an appropriate key is used in the extraction algorithm, $\bar{W} = W$, i.e. the correct watermark-embedded bits are extracted. However, if the watermarked image has been changed, or an incorrect key is used in the extraction algorithm, the extracted watermark bitmap will be visually damaged.

3 Image Quality and Authentication

The visual quality of the watermarked image can be evaluated by assessing its peak signal-to-noise ratio (PSNR). Fig. 4a shows a watermarked image with a PSNR of 37.0535 dB for a CR (compression ratio) of 13.2. Fig. 3b shows the watermark extracted from Fig. 3a. An inspection of the watermark confirms that it is intact and identical to the original watermark (not shown). The simulation results reveal that the watermarking process degrades the quality of the four original images considered in this study by no more than 0.4 dB, and hence the requirement for perceptual transparency is satisfied. Fig. 3c shows

a tampered version of Fig. 3a, in which the upper region of the original image is cropped and the area within the white rectangle is replaced by the corresponding portion of the original image. The extracted watermark presented in Fig. 3d not only provides evidence of tampering, but also indicates the particular regions of the image which have been illegally modified.

4 Recompression results

To evaluate the recompression ability of the proposed scheme, the compression ratio improvement (CRI) is computed using the equation

$$CRI = \frac{S_N - S_W}{S_N} \times 100\% \quad (3)$$

where S_N and S_W are the file sizes of the non-watermarked image and the watermarked image, respectively. Table 1 compares the sizes of the watermarked and non-watermarked versions of each of the current still images, and indicates the corresponding CRI value. It is observed that the average overall improvement in the compression ratio is approximately 5%.

5 Conclusion

This paper has presented a novel fragile compression-watermarking algorithm for the authentication of JPEG images. The simulation results have confirmed the ability of the proposed scheme to detect and localize tampering modifications. Furthermore, the results have shown that the use of a run-length selection strategy to determine the specific coefficient at which to embed the watermarked bit yields an improvement of approximately 5% in the file compression ratio.

References

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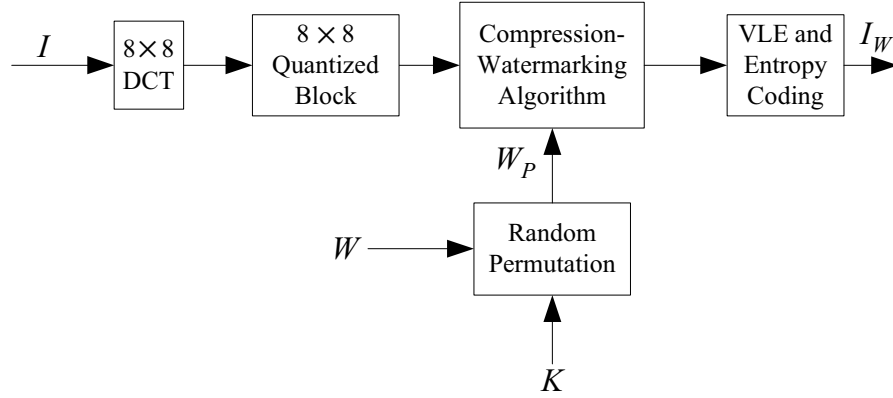


Figure 1. Schematic illustration of proposed watermark insertion procedure.



Fig. 2 Use of compression-watermarking algorithm to embed watermark in DCT-block

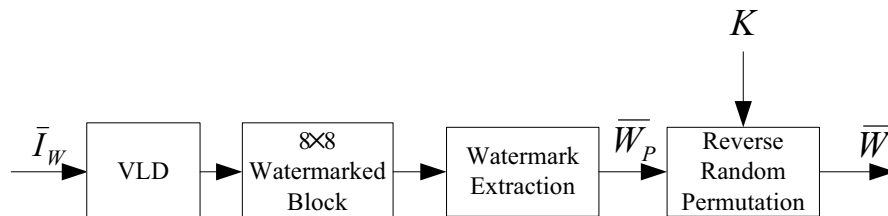


Fig. 3 Schematic illustration of proposed watermark extraction procedure.

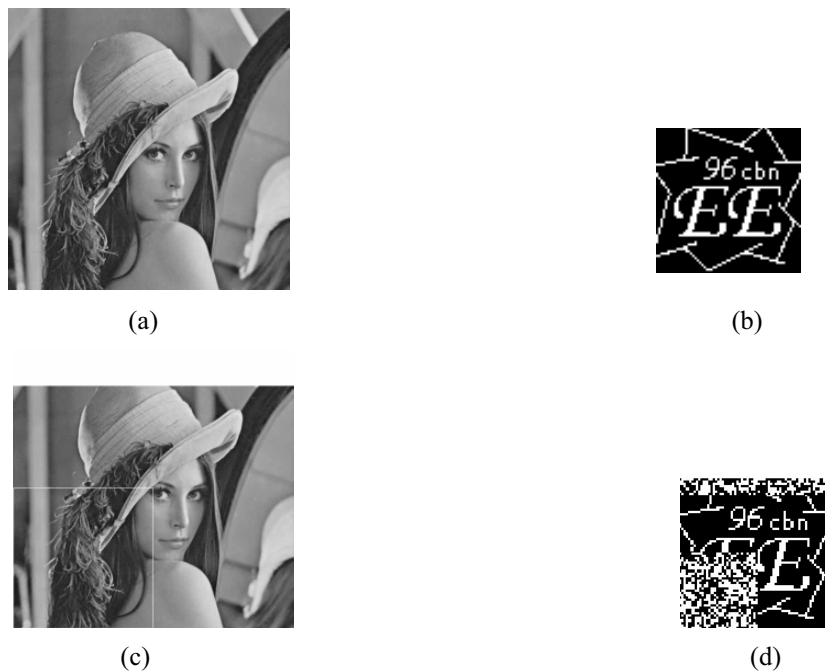


Fig. 4 (a) Watermarked image with PSNR=37.0535 at compression ratio of 13.2.
(b) Extracted watermark from Fig. 4(a).
(c) Tampering attack.
(d) Extracted watermark from Fig. 4(c).